

Task 1:

For the Titanic challenge (<https://www.kaggle.com/c/titanic>), we need to guess whether the individuals from the test dataset had survived or not.

Google Colab Code Link:

https://colab.research.google.com/drive/1YA2lzYs9T2L_HkO8cPCmxFn2VKDPheUu?usp=sharing

1. Preprocess your Titanic training data.

Answer:

The dataset was preprocessed using a scikit-learn pipeline with the following steps:

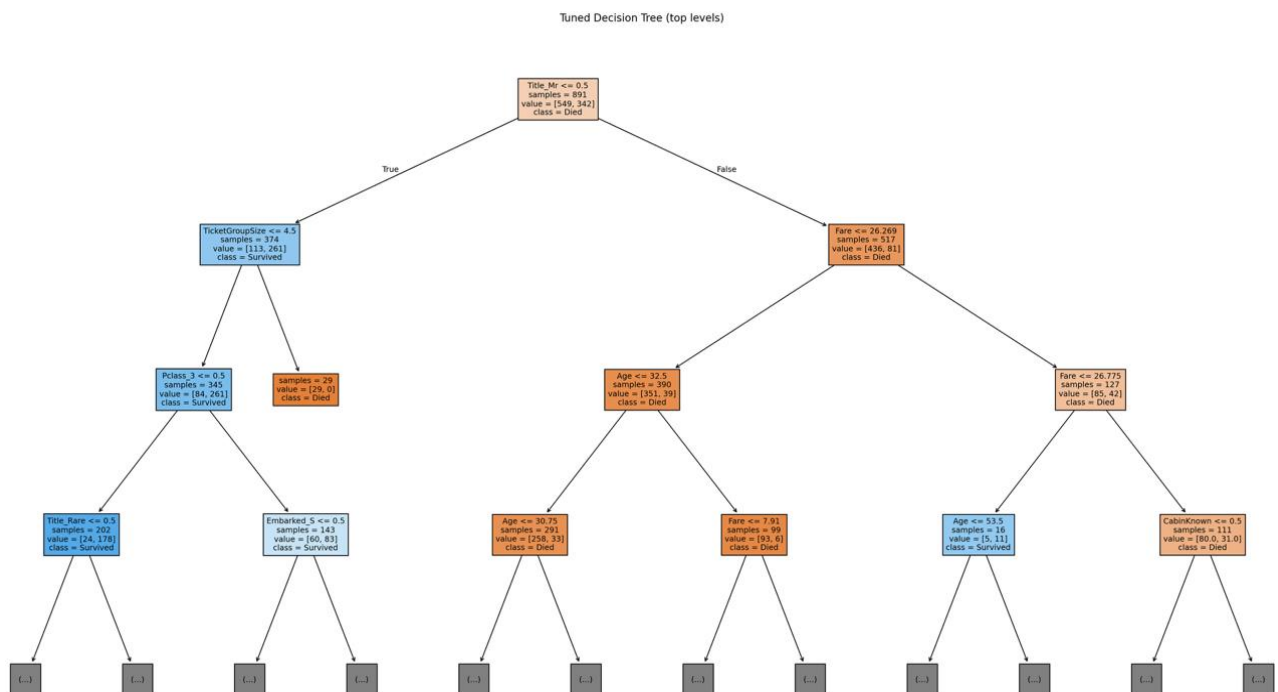
- Extracted Title from passenger names and grouped rare titles.
- Created new features: FamilySize = SibSp + Parch + 1, IsAlone (whether the passenger was traveling alone), CabinKnown (whether the cabin number was available), and TicketGroupSize (number of people sharing the same ticket).
- Imputed missing numeric values (e.g., Age, Fare) using the median, and categorical values (e.g., Embarked) using the most frequent value.
- Applied one-hot encoding to categorical variables (Sex, Embarked, Pclass, Title)
- Combined all transformations using a ColumnTransformer for consistent preprocessing during both training and testing.

2. Learn and fine-tune a decision tree model with the Titanic training data, plot your decision tree.

Answer:

A **Decision Tree Classifier** was trained and fine-tuned using Grid Search over hyperparameters such as criterion, max_depth, min_samples_split, min_samples_leaf, and ccp_alpha.

Best parameters: 'criterion': 'gini', 'max_depth': 9, 'min_samples_split': 10, 'min_samples_leaf': 1, 'ccp_alpha': 0.0



3. Apply the five-fold cross validation of your fine-tuned decision tree learning model to the Titanic training data to extract average classification accuracy.

Answer:

Five-fold cross-validation was applied to evaluate model generalization.

Average accuracy: **0.8294**

This indicates that the decision tree achieved around 82.9% accuracy on average, with small variation across folds, suggesting stable performance.

4. Apply the five-fold cross validation of your fine-tuned random forest learning model to the Titanic training data to extract average classification accuracy.

Answer:

A **Random Forest Classifier** was similarly fine-tuned with Grid Search over parameters such as n_estimators, max_depth, min_samples_split, min_samples_leaf, and max_features.

Best parameters: 'n_estimators': 600, 'max_depth': 10, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_features': None, 'bootstrap': True

Average 5-fold CV accuracy: **0.8428**

5. Which algorithm is better, Decision Tree or Random Forest? What are your observations and conclusions from the algorithm comparison and analysis?

Answer:

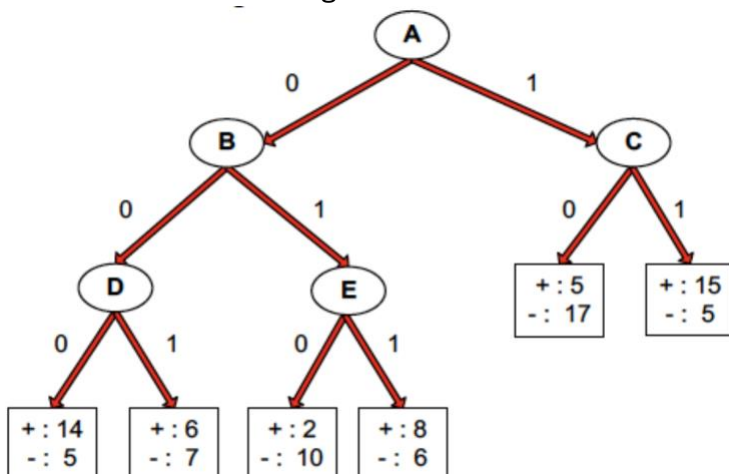
Model	Mean 5-Fold Accuracy	Std. Dev.	Remarks
Decision Tree	0.8294	0.0282	Good baseline model
Random Forest	0.8428	0.0265	Better accuracy and lower variance

Conclusion:

- The Random Forest outperformed the Decision Tree in both average accuracy and stability.
- This improvement is due to **ensemble averaging**, which reduces overfitting and variance by combining multiple trees.
- Therefore, the **Random Forest** is the preferred model for predicting Titanic survival in this task.

Task 2: Understanding Training Error and Testing

Consider the decision tree shown in the diagram below. The counts shown in the leaf nodes correspond to the number of training records associated with the nodes.



Q1: What is the training error rate for the tree? Explain how you get the answer?

Answer:

Total training samples = $(14+5) + (6+7) + (2+10) + (8+6) + (5+17) + (15+5) = 19 + 13 + 12 + 14 + 22 + 20 = 100$

Leaf	+ Count	-Count	Majority Class	Misclassified
D=0 -> +14:-5	14	5	+	5
D=1 -> +6:-7	6	7	-	6
E=0 -> +2:-10	2	10	-	2
E=1 -> +8:-6	8	6	+	6
C=0 -> +5:-17	5	17	-	5
C=1 -> +15:-5	15	5	+	5

Each leaf predicts its majority class, and misclassifications are the minority counts at each leaf.

So, total misclassified = $5 + 6 + 2 + 6 + 5 + 5 = 29$

Training error rate = $29/100 = 0.29 = 29\%$

Q2: Given a test instance $T = \{A=0, B=1, C=1, D=1, E=0\}$, what class would the decision tree above assign to T? Explain how you get the answer?

Answer:

1. $A=0 \rightarrow$ go to B
2. $B=1 \rightarrow$ go to E
3. $E=0 \rightarrow$ go to leaf with $(+2, -10)$
4. Majority = “-” $\rightarrow$ predicted class = negative

Therefore, the predicted class for T is **NEGATIVE (-)**.

Task 3: Understand Splitting Process

Consider the following data set for a binary class problem.

A	B	Class Label
T	F	+
T	T	+
T	T	+
T	F	-
T	T	+
F	F	-
F	F	-
F	F	-
T	T	-
T	F	-

Q1: What is the overall gini before splitting?

Answer:

Class counts: Total = 10, + = 4, - = 6

Overall Gini (before split) $G = 1 - ((4/10)^2 + (6/10)^2) = 0.48$

Q2: What is the gain in gini after splitting on A?

Answer:

Gain after splitting on A:

A = T (7 rows): + = 4, - = 3

$$G_{A=T} = 1 - ((4/7)^2 + (3/7)^2) = 24/49 = 0.49$$

A = F (3 rows): + = 0, - = 3

$$G_{A=F} = 1 - ((0/3)^2 + (3/3)^2) = 1 - (0+1) = 0$$

Weighted post-split Gini:

$$G_{\text{split}}(A) = (7/10)(24/49) + (3/10)(0) = 0.3429$$

$$\text{Gain in Gini (A)} = 0.48 - 0.3429 = \mathbf{0.1371}$$

Q3: What is the gain in gini after splitting on B?

Answer:

Gain after splitting on B:

B = T (4 rows): + = 3, - = 1

$$G_{B=T} = 1 - ((3/4)^2 + (1/4)^2) = 6/16 = 0.375$$

B = F (6 rows): + = 1, - = 5

$$G_{B=F} = 1 - ((1/6)^2 + (5/6)^2) = 10/36 = 0.2778$$

Weighted post-split Gini:

$$G_{\text{split}}(B) = (4/10)(0.375) + (6/10)(0.2778) = 0.31667$$

$$\text{Gain in Gini (B)} = 0.48 - 0.31667 = \mathbf{0.1633}$$

Q4: Which attribute would the decision tree choose?

Answer:

The split with the **larger Gini gain** -> B (0.1633) would be chosen.

Task 4: Please answer and explain.

Q1: Are decision trees a linear classifier? Why?

Answer:

No, decision trees are **not linear classifiers**. Linear classifiers, like logistic regression, separate classes using a single straight line (or hyperplane) in the feature space. In contrast, decision trees split the data into regions using a series of if-then rules based on individual features. These rules create **nonlinear and piecewise decision boundaries**, which allow trees to model complex patterns that linear classifiers cannot capture.

Q2: Is Misclassification error better than Gini index as the splitting criteria for decision trees? Why?

Answer:

The **Gini index** is generally preferred over the **misclassification error** as a splitting criterion. While both measure node impurity, the Gini index is more sensitive to changes in class distribution within the node. This means it better distinguishes between splits that lead to purer child nodes, resulting in more accurate and deeper trees. Misclassification error, on the other hand, is **less responsive to subtle differences** in purity, often producing less optimal splits and simpler, less accurate trees.

Task 5: What are the weaknesses of bagging? What is the difference between bagging and random forests, and why such difference can overcome the weaknesses of bagging?

Answer:

Weaknesses of Bagging:

- **High correlation between trees:** Since each tree in bagging is built using all available features, the trees can end up being similar, reducing the benefit of averaging.

- **Limited variance reduction:** If the individual trees are highly correlated, bagging won't significantly lower the overall model variance.
- **Less interpretability:** Combining many full trees makes it hard to interpret the final model.

Difference between Bagging and Random Forests:

Aspect	Bagging	Random Forest
Sampling	Bootstrap sampling of data points	Bootstrap sampling of data points and random sampling of features at each split
Feature selection at each split	Uses all features	Uses a random subset of features
Correlation among trees	Trees can be highly correlated	Feature randomness reduces correlation
Bias	Similar to single trees	Slightly higher bias due to feature randomness
Variance	Reduced by averaging, but limited by correlation	Further reduced due to lower correlation

Why Random Forests Overcome Bagging's Weakness:

- Random forests tackle bagging's **main issue**: correlated base learners.
- By choosing a **random subset of features at each node**, trees become more diverse.
- This **decorrelates** the trees, so their errors are less likely to overlap.
- Averaging less-correlated trees leads to **greater variance reduction** and **better generalization**.
- As a result, random forests achieve **lower overall error**, even if they slightly increase bias.

Task 6: Construct a support vector machine that computes the kernel function. Use four values of +1 and -1 for both inputs and outputs:

$[-1, -1]$ (negative)

$[-1, +1]$ (positive)

$[+1, -1]$ (positive)

$[+1, +1]$ (negative).

Map the input $[x_1, x_2]$ into a space consisting of x_1 and x_1x_2 . Draw the four input points in this space, and the maximal margin separator. What is the margin? [To be consistent with our lecture notes, margin is defined as the distance from the middle way/hyperplane to either support vectors.]

Answer:

Task 6:

1. Feature map and kernel function

Feature map,

$$\phi(x) = \begin{bmatrix} x_1 \\ x_1 x_2 \end{bmatrix},$$

The corresponding kernel function is

$$K(x, z) = \phi(x)^T \phi(z) = x_1 z_1 + (x_1 x_2)(z_1 z_2) = x_1 z_1 (1 + x_2 z_2)$$

2. Mapped Data Points

Original Input	Label	Mapped $(x_1, x_1 x_2)$
$(-1, -1)$	-1	$(-1, +1)$
$(-1, +1)$	+1	$(-1, -1)$
$(+1, -1)$	+1	$(+1, -1)$
$(+1, +1)$	-1	$(+1, +1)$

In this feature space, the points form a square.

- Negatives on the top row ($x_1 x_2 = +1$)
- Positives on the bottom row ($x_1 x_2 = -1$)

3. Maximal-Margin Separator.

The best separating hyperplane is:

$$x_1 x_2 = 0$$

A horizontal line in the mapped space (denoted $v=0$ if $\phi(x)=(u,v)$).

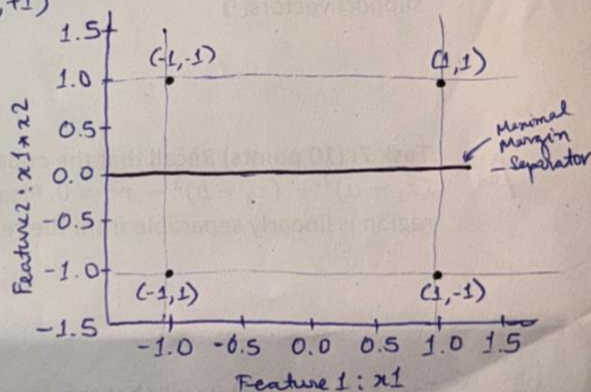
A valid SVM solution is:

$$w = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, b = 0$$

Decision function:

$$f(x) = w^T \phi(x) + b = -x_1 x_2$$

All 4 points satisfy $y_i (w^T \phi(x_i) + b) = 1$ so all are support vectors.



4. Margin Calculation

$$\text{margin} = \frac{1}{\|w\|} = \frac{1}{\sqrt{0^2 + (-1)^2}} = 1$$

Task 7: Recall that the equation of the circle in the 2-dimensional plane is $(x_1 - a)^2 + (x_2 - b)^2 - r^2 = 0$. Please expand out the formula and show that every circular region is linearly separable from the rest of the plane in the feature space (x_1, x_2, x_1^2, x_2^2) .

Answer:

Task 7:

Expand the circle equation:

$$\begin{aligned}(x_1 - a)^2 + (x_2 - b)^2 - r^2 &= 0 \\ \Rightarrow (x_1^2 - 2ax_1 + a^2) + (x_2^2 - 2bx_2 + b^2) - r^2 &= 0 \\ \Rightarrow x_1^2 + x_2^2 - 2ax_1 - 2bx_2 + (a^2 + b^2 - r^2) &= 0 \\ \Rightarrow (-2a)x_1 + (-2b)x_2 + (1)x_1^2 + (1)x_2^2 + (a^2 + b^2 - r^2) &= 0\end{aligned}$$

Map to the Feature Space

New feature space where each point (x_1, x_2) is mapped to a new point in a 4-dimensional space.

$$z_1 = x_1, z_2 = x_2, z_3 = x_1^2, z_4 = x_2^2$$

Substituting these new coordinates into our rearranged equation:

$$(-2a)z_1 + (-2b)z_2 + (1)z_3 + (1)z_4 + (a^2 + b^2 - r^2) = 0$$

Linear Separability

A linear equation defines a hyperplane. We can write the equation above in the standard form for a hyperplane, which is $w \cdot z + c = 0$, where w is a weight vector and c is a bias term.

In our case:

$$\begin{aligned}1) \text{ The weight vector is } w &= \begin{pmatrix} -2a \\ -2b \\ 1 \\ 1 \end{pmatrix} \\ 2) \text{ The feature vector is } z &= \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_1^2 \\ x_2^2 \end{pmatrix}\end{aligned}$$

$$3) \text{ The bias term is } c = a^2 + b^2 - r^2$$

The equation of the circle in the original space has been transformed into the equation of a hyperplane in the 4D feature space. This hyperplane divides the 4D space into two halves:

1. Points inside the circle, where $(x_1 - a)^2 + (x_2 - b)^2 - r^2 < 0$, now satisfy $w \cdot z + c < 0$
 2. Points outside the circle, where $(x_1 - a)^2 + (x_2 - b)^2 - r^2 > 0$, now satisfy $w \cdot z + c > 0$
- Therefore, the circular region is linearly separable by a single hyperplane in that feature space.

Task 8: Recall that the equation of an ellipse in the 2-dimensional plane is $c(x_1 - a)^2 + d(x_2 - b)^2 - 1 = 0$. Please show that an SVM using the polynomial kernel of degree 2, $K(u, v) = (1 + u \cdot v)^2$, is equivalent to a linear SVM in the feature space $(1, x_1, x_2, x_1^2, x_2^2, x_1x_2)$ and hence that SVMs with this kernel can separate any elliptic region from the rest of the plane.

Answer:

Task 8:

The Polynomial Kernel and its Feature Space

Let our input vectors be $u = (u_1, u_2)$ and $v = (v_1, v_2)$.

$$K(u, v) = (1 + u \cdot v)^2$$

Here,

$$u \cdot v = u_1 v_1 + u_2 v_2$$

$$\text{So, } K(u, v) = (1 + u_1 v_1 + u_2 v_2)^2$$

$$K(u, v) = 1 + u_1^2 v_1^2 + u_2^2 v_2^2 + 2u_1 v_1 + 2u_2 v_2 + 2u_1 u_2 v_1 v_2$$

$$K(u, v) = (1)(1) + (\sqrt{2}u_1)(\sqrt{2}v_1) + (\sqrt{2}u_2)(\sqrt{2}v_2) + (u_1^2)(v_1^2) + (u_2^2)(v_2^2) + (\sqrt{2}u_1 u_2)(\sqrt{2}v_1 v_2)$$

This shows that the kernel computation is equivalent to a dot product in a 6-D feature space. The feature map $\phi(x)$ for a vector $x = (x_1, x_2)$ is:

$$\phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1 x_2)$$

The feature space mentioned $(1, x_1, x_2, x_1^2, x_2^2, x_1 x_2)$ is a simplified version created by the kernel. An SVM using these features is functionally equivalent to one using the kernel.

Therefore, an SVM using the polynomial kernel $K(u, v) = (1 + u \cdot v)^2$ is equivalent to a linear SVM operating in this higher-dimensional feature space.

Task 8 (contd.):

We will show that the boundary of an ellipse becomes a linear separator (a hyperplane) in this new feature space.

- Ellipse equation: $c(x_1 - a)^2 + d(x_2 - b)^2 - 1 = 0$

$$c(x_1^2 - 2ax_1 + a^2) + d(x_2^2 - 2bx_2 + b^2) - 1 = 0$$

$$cx_1^2 - 2acx_1 + a^2c + dx_2^2 - 2bdx_2 + b^2d - 1 = 0$$

- Grouping terms according to the components of our feature space:

$$(c)x_1^2 + (d)x_2^2 + (-2ac)x_1 + (-2bd)x_2 + (0)x_1x_2 + (a^2c + b^2d - 1) = 0$$

This equation is a linear combination of the features in our new space.

- We can express this as a dot product $w \cdot z + b' = 0$, which is the equation for a hyperplane. Let the feature vector be $z = (x_1, x_2, x_1^2, x_2^2, x_1x_2)$. Then the equation for the ellipse becomes a linear equation with:

$$\text{Weight vector } w: w = (-2ac, -2bd, c, d, 0)$$

$$\text{Bias term } b': b' = a^2c + b^2d - 1$$

This hyperplane divides the feature space into 2 regions:

- 1) Points inside the ellipse satisfy $c(x_1 - a)^2 + d(x_2 - b)^2 - 1 < 0$, which corresponds to $w \cdot z + b' < 0$
- 2) Points outside the ellipse satisfy $c(x_1 - a)^2 + d(x_2 - b)^2 - 1 > 0$, which corresponds to $w \cdot z + b' > 0$.

Since the elliptic region can be separated from the rest of the plane by a single hyperplane in the feature space, we have shown that SVMs with a polynomial kernel of degree 2 can separate any elliptic region.

Google Colab Code Link for Task 1:

https://colab.research.google.com/drive/1YA2lzYs9T2L_HkO8cPCmxFn2VKDPheUu?usp=sharing

Additional Results for Task 1:

Decision Tree Plot: https://drive.google.com/file/d/12ZoUaY_FWR-YXWB9E1vIVb4jnLx8iZj1/view?usp=sharing

CV_Results.txt: https://drive.google.com/file/d/1pbi_84bF-ra4mMoqy31qF8GVKI8uCRSJ/view?usp=sharing

Submission.csv: <https://drive.google.com/file/d/1l-xz89mrCO9Qf1-EdAgnpSecPT-XGjls/view?usp=sharing>